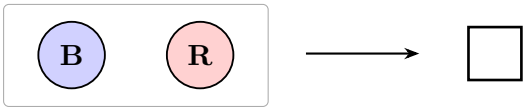
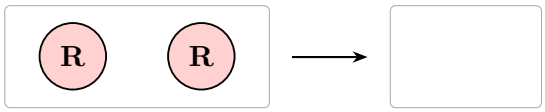
Use the same bag of three red counters and one blue. Draw twice in order, independently, giving every counter the same chance each time. Return and mix after each draw.

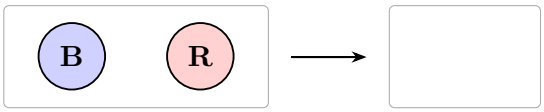


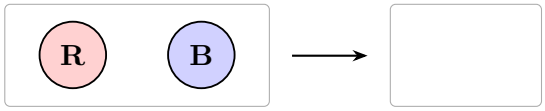
A sample rule: BR gives a square.

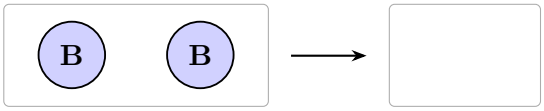
A rule gives each color pair a square, a circle, or no shape. A fair rule must sometimes give each shape, and give them equal chances when a shape comes out.

Problem 1: Design a fair rule for this bag. It must use only the colors and their order, and it must sometimes give each shape.

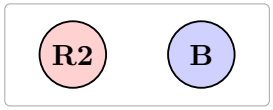








Problem 2: Check your rule using every equally likely pair of the labeled counters R1, R2, R3, B. Explain why the shape chances match.



R2 then B is one labeled pair.